

Sasan Liaw

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Education

University of California, Berkeley

Graduating May 2028

Bachelor of Science in Electrical Engineering and Computer Sciences

GPA: 3.67

Coursework: Linear ICs, Digital Design and ICs, Application-Specific ICs, Computer Architecture, Signals and Systems, Designing Information Devices and Systems, Structure and Interpretation of Computer Programs

Technical Skills

Design and Analysis: Cadence Virtuoso, Altium Designer, KiCad, Ansys HFSS, LTspice, Autodesk Fusion 360.

Programming: Java, Python, Git, RISC-V. Familiar with C, C++, SQL, LaTeX, Scheme.

Fabrication & Testing: Soldering, Solder Reflow, Electronic Testing Equipment (Oscilloscope, Adjustable Power Supply, Function Generator, and WaveForms Network Analyzer), 3D Printing, Laser Cutting.

Experience

Marvell Nanofabrication Laboratory

Berkeley, CA

Process Engineering Intern

January 2026 - Present

- Develop micron and sub-micron scale photolithography process specifications, achieving optimal feature resolution on SiO₂ and metal thin films (Al, W, Ti) using i-line photoresists (MiR701, 12XT).
- Quantitatively verify photoresist exposure quality through optical and scanning electron microscopy.

Space Enterprise at Berkeley (Liquid Rocketry Club)

Berkeley, CA

Avionics Engineer and Mentor

September 2024 - Present

- Design PCBs that process sensor data, manage power, and interface with the engine test stand architecture, including a 16-channel thermocouple amplifier and a scalable Hall effect-based fuel level tracker.
- Lead a semesterly 8-week class on circuit analysis, schematic capture, and layout, guiding students through their first PCB design project from component choice to manufacturing and assembly.
- Integrated wire harnesses, debugged DAQ PCBs, and analyzed telemetry over 500+ hours of onsite testing.
- Wrote custom firmware that executed automated calibration procedures and decoded raw sensor data.
- Collaborated with mechanical teams to seamlessly integrate sensor modules into the liquid propulsion system.

Theoretical and Applied Fluid Dynamics Laboratory (TAFLab)

Berkeley, CA

Undergraduate Researcher

August 2025 - July 2026

- Designed PCBs that manage power distribution and data acquisition for autonomous ocean-monitoring drones.
- Characterized solar panel power generation, validating its ability to sustain the system's average power draw.
- Implemented multi-rail power distribution (dropping 12V to 8V, 5V, and 3.3V) using LM2596 buck converters, optimizing layout by minimizing switching loops, isolating feedback paths, and utilizing large copper planes.
- Designed for testability, strategically placing test points, indicator LEDs, and sensor bypass pins.

Projects

Display Driver Operational Amplifier

April - May 2026

- Designed a telescopic differential input amplifier with a Class AB output stage to drive a 272×340 LCD at 60Hz using a single voltage supply and current reference, achieving 1.4V output swing in 120ns with <0.2% error.
- Validated the design in Cadence Virtuoso, iteratively improving power consumption, CMRR, and phase margin.
- Balanced trade-offs to meet all specifications, leveraging custom scripts to calculate transistor sizing and biasing conditions using gm/Id design methodology.

Magnetic Fuel Level Sensing

October 2024 - Present

- Engineered and fabricated a modular tracking device using Hall effect sensors, 8-channel ADCs, passive filtering, and a SPI bus, achieving ±0.25 cm positional accuracy with scalable architecture.
- Designed the schematic and layout of a sensor module that can interface with identical boards to create a sensor array of adjustable length while meeting a strict 11 mm maximum thickness constraint.
- Improving the device's reliability as a fuel level tracker following its integration into a rocket propellant tank.